TWO GENERALIZATIONS OF THE $\mathbb{G}_0$ DICHOTOMY

B. MILLER

Abstract. We give classical proofs of the generalizations of the $\mathbb{G}_0$ dichotomy to κ-Souslin digraphs and uniformly analytic sequences of digraphs indexed by $\kappa^{\mathbb{N}}$.

Introduction

Endow each infinite cardinal κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. We say that $\phi\colon C\subseteq\kappa^{\mathbb{N}}\times\mathbb{N}^{\mathbb{N}}\to\kappa^{\mathbb{N}}\times X$ is *parameter preserving* if $\mathrm{proj}_{\kappa^{\mathbb{N}}}\circ\phi=\mathrm{proj}_{\kappa^{\mathbb{N}}}\restriction C$, and $A\subseteq\kappa^{\mathbb{N}}\times X$ has *uniformly analytic sections* if there are a closed set $C\subseteq\kappa^{\mathbb{N}}\times\mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving map $\phi\colon C\to\kappa^{\mathbb{N}}\times X$ with $A=\phi(C)$.

A sequence $(X_i)_{i\in I}$ is *open separated* if there are pairwise disjoint open sets U_i such that $\forall i\in I\ X_i\subseteq U_i$. A subset of a topological space is *quasi-Borel* if it is in the closure of the open sets under complements, countable unions, and open-separated unions. A map $c\colon\kappa^{\mathbb{N}}\times X\to\mathbb{N}$ is *quasi-Borel* if $\forall n\in\mathbb{N}\ c^{-1}(\{n\})$ is quasi-Borel.

A *digraph* on X is an irreflexive set $G\subseteq X\times X$. A *homomorphism* from G to a digraph H on Y is a function $\pi\colon X\to Y$ with the property that $(\pi\times\pi)(G)\subseteq H$. A *$Y$-coloring* of G is a homomorphism $c\colon X\to Y$ from G to inequality. Fix $\mathbb{s}\in\prod_{n\in\mathbb{N}}2^n$ such that $\forall s\in 2^{<\mathbb{N}}\exists n\in\mathbb{N}\ s\sqsubseteq\mathbb{s}_n$ and define $\mathbb{G}_0=\bigcup_{n\in\mathbb{N}}\{(\mathbb{s}_n\frown(i)\frown c)_{i<2}\mid c\in 2^{\mathbb{N}}\}$.

The $\mathbb{G}_0$ dichotomy for uniformly analytic sequences of digraphs indexed by $\mathbb{N}^{\mathbb{N}}$ is a straightforward corollary of the $\mathbb{G}_0$ dichotomy itself (see [Mil12, Theorem 31]). We establish the following generalization to $\kappa^{\mathbb{N}}$ by modifying the classical proof of the $\mathbb{G}_0$ dichotomy:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G\subseteq\kappa^{\mathbb{N}}\times(X\times X)$ has uniformly analytic sections that are digraphs on X. Then exactly one of the following holds:*

1. *There is a quasi-Borel function $c\colon\kappa^{\mathbb{N}}\times X\to\mathbb{N}$ with the property that $\forall\sigma\in\kappa^{\mathbb{N}}\ c_\sigma$ is a coloring of G_σ.*
2. *There exists $\sigma\in\kappa^{\mathbb{N}}$ for which there is a continuous homomorphism $\pi\colon 2^{\mathbb{N}}\to X$ from $\mathbb{G}_0$ to G_σ.*

A subset of a topological space is *κ-Souslin* if it is a continuous image of a closed subset of $\kappa^{\mathbb{N}}$. A subset of a topological space is *κ^+-Borel* if

it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ. A function $c\colon X \to \kappa$ is *κ^+-Borel* if $\forall \alpha < \kappa\ c^{-1}(\{\alpha\})$ is κ^+-Borel. By combining Theorem 1 with a Baire category argument, we also show the following:

Theorem 2 (Kanovei [Kan97])**.** *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ-Souslin digraph on X. Then at least one of the following holds:*

(1) *There is a κ^+-Borel coloring $c\colon X \to \kappa$ of G.*
(2) *There is a continuous homomorphism from $\mathbb{G}_0$ to G.*

In §1, we note closure properties of sets with uniformly analytic sections. In §2, we generalize Lusin's separation theorem to sets with uniformly analytic sections. In §3, we prove Theorem 1. And in §4, we prove Theorem 2.

1. Pointclasses

We direct the reader to [Kec95, §14] for the basics on analytic sets.

Proposition 1.1. *Suppose that κ is an infinite cardinal and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proof. By the proof of the analogous fact for analytic sets. ⊠

Proposition 1.2. *Suppose that κ is an infinite cardinal, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections, and $f\colon A \to \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.*

Proof. By the proof of the analogous fact for analytic sets. ⊠

For all $s \in \kappa^{<\mathbb{N}}$, define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$.

Proposition 1.3. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $A_s \subseteq \mathcal{N}_s \times X$ has uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$. Then $\bigcup_{s \in \kappa^{<\mathbb{N}}} A_s$ has uniformly analytic sections.*

Proof. Fix closed sets $C_s \subseteq \mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s\colon C_s \to \kappa^{\mathbb{N}} \times X$ such that $A_s = \phi_s(C_s)$. If $n \in \mathbb{N}$, then $C_n = \bigcup_{s \in \kappa^n} C_s$ is closed and $\phi_n = \bigcup_{s \in \kappa^n} \phi_s$ is continuous, so $\bigcup_{n \in \mathbb{N}} \phi_n(C_n)$ has uniformly analytic sections by Proposition 1.1. ⊠

Proposition 1.4. *Suppose that κ is an infinite cardinal. Then every quasi-Borel subset of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ has uniformly analytic sections.*

Proof. The closed subsets of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ clearly have uniformly analytic sections. As metrizability ensures that the open sets are countable unions of closed sets, Proposition 1.1 ensures that every open subset of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ has uniformly analytic sections.

By Proposition 1.1, it is sufficient to show that if $(U_i)_{i\in I}$ is a pairwise disjoint sequence of open subsets of $\kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and $B_i \subseteq U_i$ has uniformly analytic sections for all $i \in I$, then so too does $B = \bigcup_{i\in I} B_i$, as this implies that if $U_i \setminus B_i$ has uniformly analytic sections for all $i \in I$, then so too does $\sim B = C \cup \bigcup_{i\in I} U_i \setminus B_i$, where $C = \sim\bigcup_{i\in I} U_i$.

But the restriction of $\mathrm{proj}_{\kappa^{<\mathbb{N}}}$ to $I_t = \{(i, s) \in I \times \kappa^{<\mathbb{N}} \mid \mathcal{N}_s \times \mathcal{N}_t \subseteq U_i\}$ is injective for all $t \in \mathbb{N}^{<\mathbb{N}}$, so Propositions 1.1 and 1.3 ensure that $B = \bigcup_{t\in\mathbb{N}^{<\mathbb{N}}} \bigcup_{(i,s)\in I_t} B_i \cap (\mathcal{N}_s \times \mathcal{N}_t)$ has uniformly analytic sections. ⊠

Proposition 1.5. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections. Then every relatively quasi-Borel subset of A has uniformly analytic sections.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi\colon C \to \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$. If $B \subseteq A$ is relatively quasi-Borel, then $\phi^{-1}(B)$ is relatively quasi-Borel in C. Propositions 1.2 and 1.4 therefore ensure that $B = \phi(\phi^{-1}(B))$ has uniformly analytic sections. ⊠

2. Separation

Recall that $B \subseteq X$ *separates* $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 2.1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a quasi-Borel set.*

Proof. Fix closed sets $C_i \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_i\colon C_i \to \kappa^{\mathbb{N}} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$, and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n\in\mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 2.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. If $i < 2$, then $(\sigma, c_i) \in C_i$, so we can set $x_i = (\mathrm{proj}_X \circ \phi_i)(\sigma, c_i)$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1. Fix $n \in \mathbb{N}$ such that $A_{i,\sigma\restriction n,c_i\restriction n} \subseteq \kappa^{\mathbb{N}} \times U_i$ for all $i < 2$. Then $A_{0,\sigma\restriction n,c_0\restriction n} \cap \overline{A_{1,\sigma\restriction n,c_1\restriction n}} = \emptyset$, which contradicts the fact that $(\sigma \restriction n, c_0 \restriction n, c_1 \restriction n) \in T$. ⊠

Set $B_{s,t,u} = {\sim}\overline{A_{1,s,u}}$ for all $(s,t,u) \in {\sim}T$ and recursively define

$$B_{s,t,u} = \bigcup_{\alpha<\kappa} (\mathcal{N}_{s^\frown(\alpha)} \times X) \cap \bigcup_{m\in\mathbb{N}} \bigcap_{n\in\mathbb{N}} B_{s^\frown(\alpha),t^\frown(m),u^\frown(n)}$$

for all $(s,t,u) \in T$. The obvious induction shows that $B_{s,t,u}$ is a quasi-Borel set separating $A_{0,s,t}$ from $A_{1,s,u}$ for all $(s,t,u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$, so $B_{\emptyset,\emptyset,\emptyset}$ is a quasi-Borel set separating A_0 from A_1. ⊠

Given $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$, we say that A is *G-independent* if $\forall \sigma \in \kappa^{\mathbb{N}}\ (A_\sigma \times A_\sigma) \cap G_\sigma = \emptyset$.

Proposition 2.3. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, the sets $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections, and A is G-independent. Then there is a quasi-Borel G-independent set $B \supseteq A$.*

Proof. By the proof of [dRM21, Proposition 1.2], using Proposition 2.1 in place of Lusin's separation theorem. ⊠

3. The uniform $\mathbb{G}_0$ dichotomy

We refer the reader to [dRM21, §1] for the classical proof of the $\mathbb{G}_0$ dichotomy that we slightly modify here.

Proof of Theorem 1. Define $p_i \colon \kappa^{\mathbb{N}} \times (X \times X) \to \kappa^{\mathbb{N}} \times X$ by setting $p_i(\sigma,(x_0,x_1)) = (\sigma,x_i)$ for all $i<2$, $\sigma \in \kappa^{\mathbb{N}}$, and $x_0,x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i<2} p_i(G)$ has uniformly analytic sections, there are closed sets $C,D \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi \colon C \to \kappa^{\mathbb{N}} \times X$ and $\psi \colon D \to \kappa^{\mathbb{N}} \times (X \times X)$ such that $\bigcup_{i<2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi^a_k)_{k<n^a})$, where $n^a \in \mathbb{N}$, $\phi^a \colon 2^{n^a} \to \mathbb{N}^{n^a}$, and $\psi^a_k \colon 2^{n^a-(k+1)} \to \mathbb{N}^{n^a}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

(1) $n^b = n^a + 1$,
(2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b}\ (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
(3) $\forall k < n^a \forall u \in 2^{n^a-(k+1)} \forall v \in 2^{n^b-(k+1)}\ (u \sqsubseteq v \implies \psi^a_k(u) \sqsubseteq \psi^b_k(v))$.

Given $\sigma \in \kappa^{\mathbb{N}}$, a *$\sigma$-configuration* is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi^\gamma_k)_{k<n^\gamma})$, where $\phi^\gamma \colon 2^{n^\gamma} \to \mathbb{N}^{\mathbb{N}}$ and $\psi^\gamma_k \colon 2^{n^\gamma-(k+1)} \to \mathbb{N}^{\mathbb{N}}$ for all $k < n^\gamma$, such that:

(1) $\forall u \in 2^{n^\gamma}\ \phi^\gamma(u) \in C_\sigma$,
(2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma-(k+1)}\ \psi^\gamma_k(v) \in D_\sigma$, and
(3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma-(k+1)}$
$(\mathrm{proj}_{X\times X} \circ \psi_\sigma \circ \psi^\gamma_k)(v) = ((\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathbb{s}_k {}^\frown (i) {}^\frown v))_{i<2}$.

We say that γ is *compatible* with $Y \subseteq X$ if $(\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

(1) $n^\gamma = n^a$,
(2) $\forall u \in 2^{n^a}\ \phi^a(u) \sqsubseteq \phi^\gamma(u)$, and
(3) $\forall k < n^a \forall v \in 2^{n^a-(k+1)}\ \psi^a_k(v) \sqsubseteq \psi^\gamma_k(v)$.

Given a set $Z \subseteq \kappa^{\mathbb{N}} \times X$, we say that a is *Z-terminal* at σ if no σ-configuration is compatible with Z_σ and a one-step extension of a. Let $A(a, Z)$ denote the set of pairs of the form $(\sigma, (\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathbb{s}_{n^a}))$, where $\sigma \in \kappa^{\mathbb{N}}$ and γ is a σ-configuration compatible with a and Z_σ.

Lemma 3.1. *Suppose that a is an approximation, $\sigma \in \kappa^{\mathbb{N}}$, $Z \subseteq \kappa^{\mathbb{N}} \times X$, and a is Z-terminal at σ. Then $A(a, Z)_\sigma$ is G_σ-independent.*

Proof. By the proof of [dRM21, Lemma 1.5]. ⊠

Lemma 3.2. *If a is an approximation and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel, then $A(a, B)$ has uniformly analytic sections.*

Proof. The set of pairs (σ, γ) for which γ is a σ-configuration compatible with a is closed, so the set of such pairs that are compatible with B_σ is quasi-Borel, thus Propositions 1.2 and 1.5 yield the desired result. ⊠

We say that a is *Z-terminal* on a set $U \subseteq \kappa^{\mathbb{N}}$ if it is Z-terminal at σ for all $\sigma \in U$. Set $A_s(a, Z) = A(a, Z) \cap (\mathcal{N}_s \times X)$. If $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel and a is B-terminal on $\mathcal{N}_s$, then Proposition 2.3 and Lemmas 3.1 and 3.2 yield a G-independent quasi-Borel set $A_s(a, B) \subseteq B_s(a, B) \subseteq \mathcal{N}_s \times X$.

Let $\mathscr{A}$ be the set of all approximations and $\mathscr{A}_n = \{a \in \mathscr{A} \mid n^a = n\}$ for $n \in \mathbb{N}$. We will construct a decreasing sequence $(B^\alpha)_{\alpha<\kappa^+}$ of quasi-Borel subsets of $\kappa^{\mathbb{N}} \times X$ and an increasing sequence $(S^\alpha)_{\alpha<\kappa^+}$ of subsets of $\bigcup_{n\in\mathbb{N}} \kappa^n \times \mathscr{A}_n$, beginning with $S^0 = \emptyset$ and setting $S^\lambda = \bigcup_{\alpha<\lambda} S^\alpha$ for all limit ordinals $\lambda < \kappa^+$. Given $\alpha < \kappa^+$, $(B^\beta)_{\beta<\alpha}$, and $(S^\beta)_{\beta\leq\alpha}$, define $\rho^\alpha \colon S^\alpha \to \alpha$ by $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$, set $B^\alpha = {\sim}\bigcup_{(s,a)\in S^\alpha} B_s(a, B^{\rho^\alpha(s,a)})$, let $S^{\alpha+1}$ be the set of $(s, a) \in \bigcup_{n\in\mathbb{N}} \kappa^n \times \mathscr{A}_n$ for which a is B^α-terminal on $\mathcal{N}_s$, and fix $B_s(a, B^\alpha)$ as above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. As $|\kappa^{<\mathbb{N}} \times \mathscr{A}| = \kappa$, there is a least $\lambda < \kappa^+$ with $S^\lambda = S^{\lambda+1}$, so $B^\lambda = B^{\lambda+1}$. Set $B = B^\lambda$, $S = S^\lambda$, and $\rho = \rho^\lambda$.

Lemma 3.3. *Suppose that $\alpha < \kappa^+$, a is an approximation, $s \in \kappa^{n^a}$, and a is not $B^{\alpha+1}$-terminal on $\mathcal{N}_s$. Then there exist $\xi < \kappa$ and a one-step extension b of a that is not B^α-terminal on $\mathcal{N}_{s^\frown(\xi)}$.*

Proof. Fix $\sigma \in \mathcal{N}_s$ for which a is not $B^{\alpha+1}$-terminal at σ. Then there is a σ-configuration γ compatible with a one-step extension b of a and $B^{\alpha+1}_\sigma$. Set $\xi = \sigma(n^a)$. If b is B^α-terminal on $\mathcal{N}_{s^\frown(\xi)}$ and $\beta = \rho^{\alpha+1}(s \frown (\xi), b)$, then $(\sigma, (\mathrm{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathbb{s}_{n^b})) \in A_{s^\frown(\xi)}(b, B^\beta) \cap B^{\alpha+1} \subseteq B_{s^\frown(\xi)}(b, B^\beta) \cap B^{\alpha+1} = \emptyset$, the desired contradiction. ⊠

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 3.4. *Suppose that a_0 is not B-terminal on $\kappa^{\mathbb{N}}$. Then there is a continuous homomorphism from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

Proof. By Lemma 3.3 there exist $a_n \in \mathscr{A}$ with $n^{a_n} = n$, $s_n \in \kappa^n$, $\sigma_n \in \mathcal{N}_{s_n}$, and a σ_n-configuration γ_n compatible with a_{n+1} and B_{σ_n}, where a_{n+1} is a one-step extension of a_n and $s_n \sqsubseteq s_{n+1}$, for all $n \in \mathbb{N}$.

Set $\sigma = \bigcup_{n\in\mathbb{N}} s_n$. Then $\sigma_n \to \sigma$. Define $\phi' \colon 2^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ and $\psi'_k \colon 2^{\mathbb{N}} \to \mathbb{N}^{\mathbb{N}}$ by $\phi'(c) = \bigcup_{n\in\mathbb{N}} \phi^{a_n}(c \restriction n)$ and $\psi'_k(c) = \bigcup_{n>k} \psi_k^{a_n}(c \restriction (n - (k+1)))$. If $c \in 2^{\mathbb{N}}$ and $k \in \mathbb{N}$, then $(\sigma_n, \phi^{\gamma_n}(c \restriction n+1)) \to (\sigma, \phi'(c))$ and $(\sigma_n, \psi_k^{\gamma_n}(c \restriction (n-k))) \to (\sigma, \psi'_k(c))$, so the fact that C and D are closed ensures that $(\sigma, \phi'(c)) \in C$ and $(\sigma, \psi'_k(c)) \in D$.

Define $\pi \colon 2^{\mathbb{N}} \to X$ by $\pi(c) = (\mathrm{proj}_X \circ \phi_\sigma)(\phi'(c))$ and observe that $(\mathrm{proj}_{X\times X} \circ \psi_\sigma)(\psi'_k(c)) = (\pi(\mathbb{s}_k \frown (i) \frown c))_{i<2}$ for all $k \in \mathbb{N}$ and $c \in 2^{\mathbb{N}}$, so π is a continuous homomorphism from $\mathbb{G}_0$ to G_σ. ⊠

By Lemma 3.4, we can assume that a_0 is B-terminal on $\kappa^{\mathbb{N}}$. Set $\alpha = \rho(\emptyset, a_0)$. Then $A(a_0, B) = B \cap \bigcup_{i<2} p_i(G)$ and $A(a_0, B^\alpha) \subseteq B_\emptyset(a_0, B^\alpha)$, so $A(a_0, B) = \emptyset$, thus B is G-independent, as are the quasi-Borel sets $B_a = \bigcup_{s\in S^a} B_s(a, B^{\rho(s,a)})$ for all $a \in \mathscr{A}$, hence (1) follows from the countability of $\mathscr{A}$ and the fact that $\kappa^{\mathbb{N}} \times X = B \cup \bigcup_{a\in\mathscr{A}} B_a$.

It only remains to note that if both (1) and (2) hold, then there is a quasi-Borel $\mathbb{N}$-coloring of $\mathbb{G}_0$, which is Borel by the second countability of $2^{\mathbb{N}}$, a contradiction (see, for example, [dRM21, Proposition 1.3]). ⊠

4. Stratifications and category transforms

We direct the reader to [Kec95, §8] for the basics on Baire category. For all $B \subseteq \kappa^{\mathbb{N}} \times X$ and $t \in \kappa^{<\mathbb{N}}$, define

$$B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}.$$

Proposition 4.1. *Suppose that κ is an infinite cardinal, $t \in \kappa^{<\mathbb{N}}$, X is κ-Souslin, and $B \subseteq \kappa^{\mathbb{N}} \times X$ is quasi-Borel. Then B^{*t} is κ^+-Borel.*

Proof. If B is open, then set $U_v = \bigcup\{U \subseteq X \text{ open} \mid \mathcal{N}_v \times U \subseteq B\}$ and note that $B^{*t} = \bigcap_{u\sqsupseteq t} \bigcup_{v\sqsupseteq u} U_v$ is κ^+-Borel. If $\forall t \in \kappa^{<\mathbb{N}}\ B^{*t}$ is κ^+-Borel, then $\forall t \in \kappa^{<\mathbb{N}}\ (\sim B)^{*t} = \sim\bigcup_{u\sqsupseteq t} B^{*u}$ is κ^+-Borel. If B is the union of a countable or open-separated sequence $(B_i)_{i\in I}$ of non-empty sets such that $\forall i \in I \forall t \in \kappa^{<\mathbb{N}}\ B_i^{*t}$ is κ^+-Borel, then $|I| \leq \kappa$ since X is κ-Souslin, so $\forall t \in \kappa^{<\mathbb{N}}\ B^{*t} = \bigcap_{u\sqsupseteq t} \bigcup_{v\sqsupseteq u} \bigcup_{i\in I} B_i^{*v}$ is κ^+-Borel. ⊠

Given a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi \colon C \to X$, define $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi' \colon C' \to \kappa^{\mathbb{N}} \times X$ by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. A *stratification* of a κ-Souslin set $A \subseteq X$ is

a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ for which there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi\colon C \to X$ such that $A = \phi(C)$ and $A' = \phi'(C')$. We write $\forall^*\sigma \in \kappa^{\mathbb{N}}\ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager.

Proposition 4.2. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, $A \subseteq X$ is κ-Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A, and $x \in A$. Then $\forall^*\sigma \in \kappa^{\mathbb{N}}\ x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi\colon C \to X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for which $\phi(\tau) = x$. Then the set $H = \bigcap_{n\in\mathbb{N}} \bigcup_{s\in\kappa^{<\mathbb{N}}} \mathcal{N}_{s^\frown(\tau(n))}$ is a dense G_δ subset of $\kappa^{\mathbb{N}}$ and $\forall\sigma \in H\ x \in A'_\sigma$. ⊠

Proposition 4.3. *Suppose that κ is an infinite cardinal, $t \in \kappa^{<\mathbb{N}}$, X is a Hausdorff space, G is a κ-Souslin digraph on X, G' is a stratification of G, and $B \subseteq \kappa^{\mathbb{N}} \times X$ is G'-independent. Then B^{*t} is G-independent.*

Proof. Otherwise, fix $x, y \in B^{*t}$ with $x\ G\ y$ and appeal to Proposition 4.2 to obtain $\sigma \in \mathcal{N}_t$ with $x, y \in B_\sigma$ and $x\ G'_\sigma\ y$, a contradiction. ⊠

Proof of Theorem 2. Fix a stratification G' of G. By Theorem 1 we can assume that there is a quasi-Borel function $c'\colon \kappa^{\mathbb{N}} \times X \to \mathbb{N}$ with the property that $\forall\sigma \in \kappa^{\mathbb{N}}\ c'_\sigma$ is a coloring of G'_σ. Note that the set $A = \{x \in X \mid \exists y \in X\ (x\ G\ y \text{ or } y\ G\ x)\}$ is κ-Souslin. For all $n \in \mathbb{N}$ and $t \in \kappa^{<\mathbb{N}}$, set $B_n = (\kappa^{\mathbb{N}} \times A) \cap c'^{-1}(\{n\})$, note that $A \cap B^{*t}_n$ is relatively κ^+-Borel in A and G-independent by Propositions 4.1 and 4.3, and observe that any κ^+-Borel set $C_{n,t} \subseteq X$ with $A \cap B^{*t}_n = A \cap C_{n,t}$ is also G-independent. To see that there is a κ^+-Borel coloring of G, it only remains to show that the set $C' = {\sim}\bigcup_{n\in\mathbb{N},t\in\kappa^{<\mathbb{N}}} C_{n,t}$ is G-independent. For this, we need only check that $A \cap C' = \emptyset$. But if $x \in A$, then $\kappa^{\mathbb{N}} = \bigcup_{n\in\mathbb{N}} B^x_n$, so there exists $n \in \mathbb{N}$ for which B^x_n is non-meager, thus there exists $t \in \kappa^{<\mathbb{N}}$ for which $x \in B^{*t}_n$, hence $x \in C_{n,t}$. ⊠

As noted in [Mil12, §8], the weakening of Theorem 2 without the definability condition on c follows from the simplification of the classical proof of the $\mathbb{G}_0$ dichotomy in which the use of separation is removed. As separation easily generalizes to κ-Souslin sets, this suggests a simpler route towards a classical proof of the full version of Theorem 2. The direct adaptation of that proof, however, would require separation for continuous images of κ^+-Borel sets. To see that this is consistently false, note that every $\mathbf{\Pi}^1_2$ subset of $\mathbb{N}^{\mathbb{N}}$ is $\aleph_2$-Borel (see, for example, [Kec95, Exercise 38.8]), so every $\mathbf{\Sigma}^1_3$ subset of $\mathbb{N}^{\mathbb{N}}$ is a continuous image of an $\aleph_2$-Borel set. Under $\mathtt{MA}+\neg\mathtt{CH}$, the sets with the Baire property are closed under $\aleph_1$-unions, so every $\aleph_2$-Borel subset of $\mathbb{N}^{\mathbb{N}}$ has the Baire property. But this is consistent with the existence of a $\mathbf{\Delta}^1_3$ set $B \subseteq \mathbb{N}^{\mathbb{N}}$

without the Baire property (see [HS85, Claim 8]), and no such B is $\aleph_2$-Borel, although B and $\sim B$ are continuous images of $\aleph_2$-Borel sets.

Acknowledgements. ChatGPT and Gemini provided substantial assistance in the discovery and development of the results presented here.

References

[dRM21] N. de Rancourt and B. D. Miller, *The Feldman–Moore, Glimm–Effros, and Lusin–Novikov theorems over quotients*, 2021.

[HS85] Leo A. Harrington and Saharon Shelah, *Some exact equiconsistency results in set theory*, Notre Dame J. Formal Logic **26** (1985), no. 2, 178–188.

[Kan97] Vladimir Kanovei, *Two dichotomy theorems on colourability of non-analytic graphs*, Fundamenta Mathematicae **154** (1997), no. 2, 183–201.

[Kec95] Alexander S. Kechris, *Classical descriptive set theory*, Graduate Texts in Mathematics, vol. 156, Springer-Verlag, New York, 1995.

[Mil12] Benjamin D. Miller, *The graph-theoretic approach to descriptive set theory*, Bull. Symbolic Logic **18** (2012), no. 4, 554–575.